

Which Robot Do You Prefer When Boarding an Elevator: Fast vs. Considerate

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Figure 1: Survey Video Clips: Robot positions of Plain Path condition (Left, 2 images) / Robot positions of Considerate Path condition (Right, 2 images)

ABSTRACT

As robots are increasingly integrated into our everyday lives, elevator situations with robots have become unavoidable. Elevators are essential facilities for moving between floors within high-rise buildings, yet they often serve as spaces that generate dissatisfaction among people. It is imperative that robots do not exacerbate this dissatisfaction among elevator users. To address this issue, we conducted an online comparative survey using video-based scenarios to investigate the impact of a robot's boarding time and boarding position on elevator satisfaction among office employees who coexist with real robots in their daily work environment ($N = 374$). The results revealed that both of these factors significantly contribute to satisfaction. Additionally, we found that the robot's perceived competence is the most influential factor for satisfaction. This case study underscores the importance of time for individuals in elevator environments within the context of Human-Robot Interaction (HRI) research and provides suggestions for enhancing people's satisfaction.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**.

KEYWORDS

Human-Robot Interaction, Elevators, Social Navigation, Robot Behaviors

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1 INTRODUCTION

The utilization of mobile robots in our daily lives is on the rise, particularly in high-rise buildings where population density is higher, thereby increasing the demand for robots. In high-rise buildings, the elevator is essential for vertical transportation of both people and goods. In such an environment, situations where robots share elevators with people are inevitable. Elevators serve as bottlenecks for intra-building mobility and are often a source of dissatisfaction among individuals. Due to the perception that robots may be less agile than humans, people anticipate increased discomfort and dissatisfaction when robots use these elevators. Indeed, we frequently receive complaints from individuals who associate extended elevator waiting times with the presence of robots, solely on the grounds that people share elevators with these machines.

In a 28-story office building where approximately 5,000 individuals are employed, we provide over 100 robot delivery services per day. Starting from the architectural design phase, we have incorporated dedicated vertical transport systems exclusively for robots to minimize people's dissatisfaction and enhance robot movement efficiency. However, in situations where there is a high demand for robot services or for testing purposes, robots may also use passenger elevators. In this context, we conducted tests to alter robot interactions and positions from a Human-Robot Interaction (HRI) perspective, aiming to reduce people's dissatisfaction and facilitate the easy interpretation of robot intentions. While conducting experiments, we referred to several prior research findings. However, applying these findings directly to our complex real-world environment proved to have limitations. During the operation of robot services, it was observed that the most significant source of dissatisfaction among people was the increase in elevator usage times. Additionally, some factors aimed at enhancing the social acceptance of robots were found to inadvertently prolong the elevator usage times. The primary factor contributing to this was the robot's

position. Due to the current limitations in robot sensing and mobility capabilities compared to humans, more time was required for robots to reach the desired positions proposed in [5]. Consequently, we aim to investigate which factor, between position and time, has a more positive impact on the elevator experience.

This study contributes to the HRI community in the following ways: Drawing from the experiences and dissatisfactions of office employees who coexist with real robots in their daily work environment, we establish priorities and research directions regarding the factors to consider when robots board elevators.

2 RELATED WORK

2.1 Elevator Experience

We initially explored the factors that influence people's satisfaction when using elevators, regardless of the presence of robots. In the elevator industry, there are well-established criteria for elevator quality improvement, including waiting time, traveling time, and intermediate stops. Ideally, it's better to reduce all three of these factors if possible, but in the real world, robots are bulkier and slower than humans, so they often contribute to increasing these factors. It is reported that people tend to be more dissatisfied when waiting for the elevator or during intermediate stops rather than during the actual travel phase [2]. The lower satisfaction with waiting time can be attributed to psychological factors, as occupied time feels shorter than unoccupied time, and people want to get started [10]. The time taken by robots to board the elevator can delay the initiation of people's vertical movement (time they start using the elevator). In addition to the time spent on passenger boarding and disembarking during intermediate stops, there may be additional delays in departure, such as waiting for someone walking from a distance or waiting for potential additional passengers [14]. In the case of robots, there may be times when they wait without any action due to communication delays or other reasons. Alongside efforts to reduce physical waiting time, adding distractors to reduce psychological waiting time is also a viable approach [11]. A prominent example of this applied in elevator environments includes mirrors or digital advertisements, which occupy waiting time with something other than emptiness. It is worth considering what the robots can offer to fill (occupy) the time that they have increased. In terms of elevator User Experience (UX), Rebekah Rousi [13] mentioned that people should not consciously experience elevator interaction. Negative experiences, such as slowness or perceived threats, become conscious experiences that people remember and recall. If robots provide negative experiences, people are more likely to remember those moments distinctly.

2.2 Studies on Elevator in HRI

In the field of HRI, there has been a recent increase in research directly focused on robots and elevators. D. Gallo et al. [4, 5] have proposed machine-like robot positioning within elevators, addressing both the waiting position and the standing position after boarding. They conducted online surveys and in-person experiments to determine suitable robot positions. In this study, they proposed the preferred positions for the robot as follows: a fixed position next to the elevator door while waiting for the elevator and a position inside the elevator near the door that does not block other

riders. Regarding priority claims for boarding, they expressed the robot's intent through motion, and it was found that most people appreciated gestures of concession. However, some exceptions aside, individuals generally indicated their preference for the robot to yield and let them board first.

Due to the limited space in the elevator setting, the robot may require permission from individuals or induce certain behaviors in order to complete its task [8]. A typical scenario is when the robot requests permission from people who have already boarded and asks them to make room [7]. Babel et al. [1] conducted a comparative evaluation of social acceptance in situations where robots potentially collide with people while using elevators. They varied robot appearance, levels of politeness, and modalities. Their findings indicated that polite appeals resulted in higher social acceptance compared to commands, but politeness alone did not necessarily lead people to follow the robot's requests.

In our literature review, we did not identify any prior research specifically examining robot-related factors associated with elevator satisfaction, particularly those related to time considerations. While there are some studies that explore robot speed and proximity in hallway environments [6, 9, 12], it seems that there is a dearth of research specifically examining the time it takes for robots to board and disembark from elevators and how it relates to the social acceptance of waiting passengers.

While it may seem obvious that faster robot movements within a safe range would generally be preferable, it is possible that this aspect remains a challenge due to the limitations of current robot capabilities. However, to effectively integrate robots into real-life scenarios, it appears necessary to assess the prioritization of various factors that influence each other, including those related to elevator interactions. Understanding the relative importance of these factors is crucial for successful robot deployment in practical settings.

3 USER EXPERIMENT

3.1 Research Objective

In our building, we adopted the robot standing positions proposed in [5]. However, we discovered limitations in robot movements while attempting to adhere to these positions seamlessly. To move to the corner of the elevator car near the door, the robot needs prior sensing of that space. Until the robot passes through the elevator doors, that space remains a blind spot, preventing the robot from moving as flexibly as people expect. Additionally, this motion sometimes caused time delays or made the robot appear less proficient.

Considering that time may be the most influential factor in elevator satisfaction, we began to question whether it would be beneficial for the robot to board as quickly as possible. To achieve rapid robot boarding, the most direct and straightforward path should be followed.

We aim to formulate hypotheses on two specific factors and conduct a survey among individuals who have encountered the robot to validate them with high confidence.

Robot Boarding Time and Elevator User Satisfaction

- H_{A0} : Robot boarding speed (time taken to board) will have no impact on elevator satisfaction.

- H_{A1} : Elevator satisfaction will be higher when accompanied by a robot that boards quickly compared to a robot that boards slowly.

Robot Waiting/Boarding Positions and Elevator User Satisfaction

- H_{B0} : Assuming equal boarding times, robot waiting/boarding positions will have no impact on elevator satisfaction.
- H_{B1} : Even with equal boarding times, robot waiting/boarding positions will influence elevator satisfaction.

For the first hypothesis, it may be natural to assume that being faster within a safe range would inherently have a positive impact on the elevator experience. However, a rapid boarding by the robot might influence its social acceptance, and this, in turn, could affect elevator satisfaction. We will thus investigate this aspect.

The second hypothesis aims to determine whether the robot's position has an impact on elevator satisfaction, even when boarding times are equal. Although these hypotheses do not explicitly mention it, we will also measure the Robotic Social Attribute Scale (RoSAS) in relation to speed and position. Particularly regarding the robot's position, we will directly compare the social acceptance between the recommended position suggested in [5] and a straightforward position chosen as the control group.

Through this experiment, we intend to create a guide for practical use, focusing on robot boarding positions and times that not only enhance satisfaction but also maintain high social acceptance among people.

3.2 Participants

We conducted a survey targeting 5,000 employees working in the building through an internal bulletin board, receiving valid responses from 374 individuals. Participants were informed that they could enter a drawing for a reward at a level of 7.5 US dollars. The reason for exclusively targeting employees was to capture the real voices of people who perceive robots as part of their daily lives, no longer as a novelty or something to be considered with care, but as a service that is integrated into their lives without causing harm.

The age of the respondents ranged from 22 to 55 years, with an average age of 34.1 years ($SD = 6.81$). Among them, there were 228 males, 145 females, and 1 individual who did not disclose their gender. Since robot services had been provided for more than a year, employees had ample opportunities to interact with robots. All participants were aware of the robot service, and among the respondents, 283 (75.7%) had directly used robot delivery services (e.g., parcel, food, or beverage delivery), and when combined with those who had indirectly observed others using robot services, the total was 344 (92.0%). Among these, 370 individuals (98.9%) had encountered a moving robot.

3.3 Robot

The robot used in the experiment was our own development currently providing services within the building. This robot has a footprint of 50×55 cm and stands at a height of 107 cm. Depending on the surrounding environment, it can move at higher speeds, but within the elevator environment, it operates at speeds of 0.7 m/s or lower, in compliance with industrial safety standards. The robot is designed with reinforcement learning-based obstacle avoidance

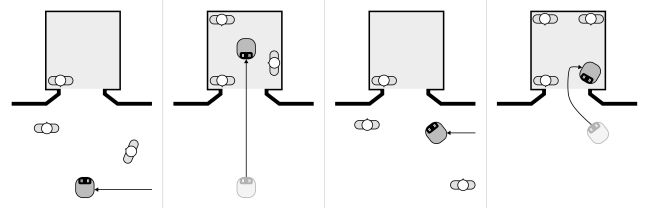


Figure 2: Robot paths and positions: Plain (PF & PS) (Left, 2 images) / Considerate (CF & CS) (Right, 2 images)

and path planning capabilities, and it immediately stops when an obstacle is detected within 20 cm.

In this experiment, to exclude the autonomy of the robot, it was manually controlled for movement, and certain video editing was performed to synchronize the timing conditions. The robot is equipped with light elements in addition to a front display; however, during the experiment, only the gaze, representing the robot's "face," was displayed without additional messages or status indicators.

3.4 Experimental Design

The study follows a 2x2 within-subjects design. The independent variables are the robot's position (1. Plain (P), 2. Considerate (C)) and boarding time (1. Fast (F, 8 seconds), 2. Slow (S, 16 seconds)), and to eliminate order effects, the four trials are provided randomly.

Boarding Time: Although the actual travel time of the robot may vary depending on the environment, the boarding times used in this survey were selected as a basis for comparison. They represent the ideal times based on a speed of 0.7 m/s, which is the robot's speed in an obstacle-free, ideal situation, taking the simplest path. The two chosen boarding times are 8 seconds, representing the fastest time achievable in an ideal environment, and 16 seconds, representing the minimum time it takes for the robot to move to the considerate position. These times were determined through multiple attempts and fine-tuning, as the actual time was significantly longer. However, for the experiment, we considered the ideal times, ensuring that even with different robot paths in the video, the same time was taken.

Robot's Position: There were two categories of robot position: "Plain" and "Considerate." "Plain" represents the robot starting 1.5 meters away from the elevator door, directly facing it. The robot moves straight through the center of the elevator door, enters about three fifth of the way into the elevator car, and then makes a 180-degree turn to face the elevator door. "Considerate" represents the robot waiting for the elevator near the right wall of the elevator door, standing at a distance of about 30 cm from the wall without blocking the door. When the elevator door opens, the robot moves straight through the center of the door and stands near the wall closest to the elevator door, opposite the wall with the floor selection buttons. This position is both machine-like and has a high level of social acceptance.

Based on research indicating that people prefer to board the elevator before the robot [4], the robot always boarded later than the people. The time it took for the robot to board after the door opened and the time for the robot to complete boarding and for the door to close were all kept consistent. To replicate actual boarding

situations, actors accompanied the robot during boarding. These actors behaved consistently with no significant impact on the experiment. Their standing positions were determined by referencing [5]. The elevator dimensions are 185 cm (width) by 200 cm (depth), with a door width of 110 cm.

3.5 Procedure

The experiment was conducted using the online survey platform surveymonkey.com from September 18, 2023, to September 21, 2023, and it typically took participants less than 10 minutes to complete the survey. To obtain multiple responses, the experiment was conducted using video-based stimuli. Survey participants observed the robot's boarding behavior from a third-person perspective through videos. These videos depicted the robot's boarding process outside the elevator door, as well as its behavior inside the elevator car, mimicking the actual scenarios experienced by employees during their daily commute.

Participants did not have direct physical interactions with the robot in a real-world setting. Instead, they watched videos featuring two actors who boarded the elevator with the robot. By observing these actors in the video, participants were indirectly prompted to imagine themselves boarding the elevator with the robot. All participants boarded the elevator from the same location and stood in the same position inside the elevator car.

Two cameras were used for recording, positioned in fixed locations both outside and inside the elevator car. The camera outside the elevator was set slightly below the average eye level, at a height of 140 cm, to capture the perspective of an observer. The camera inside the elevator car was placed at a height of 2.2 meters to ensure an adequate view of the robot's position and a partial view of the floor.

All videos begin with a message stating, "Waiting for the elevator to attend a meeting on a different floor in 1 minute," creating a scenario where survey participants are in a hurry. As participants wait for the elevator, a robot appears to board the same elevator. The robot stands at one of the two different positions defined earlier, either "Plain" or "considerate." After a brief moment, the elevator doors open, and people board the elevator. After people have boarded, the robot also boards the elevator. The video perspective switches to the camera inside the elevator as the robot passes through the elevator doors. The robot stops at one of the two different positions defined earlier, changes direction, and the elevator doors close. Depending on the scenario, the robot takes different amounts of time to travel from its starting position to its final stopping point, either 8 seconds or 16 seconds, along the same path.

The duration of the videos is 32 seconds for the "fast" scenario and 40 seconds for the "slow" scenario. In all videos, there is an 11-second wait after the video starts until the elevator doors open. The time at which the doors open, and the robot starts boarding is 15 seconds after the video begins. Except for the boarding time, the remaining time for the doors opening and closing is the same.

Participants watch these scenarios through videos and then answer questions based on their observations. To eliminate sequence effects, the four trials are presented in a random order.

3.6 Questionnaire

All survey participants are provided with an explanation of the survey's purpose and the handling of personal information before responding. Before evaluating the different cases, participants are asked about their prior experiences with robots and their level of knowledge. For each case (trial), participants watch the video and provide responses using a Likert scale 9-point rating on emotional satisfaction as follows:

"How would you feel when using the elevator with this robot? (1: 😞 - 5: 😐 (Neutral) - 9: 😊)"

Elevator satisfaction is assessed using emoticons to distinguish between positive and negative sentiments rather than selecting specific words. In addition to the 9-point scale, participants can provide comments on their reasons. Subsequently, they were asked to evaluate the 18 items presented in the RoSAS [3] using a 9-point scale. Finally, participants rate how they perceived the robot's movement speed in the video relative to their expectations using a 9-point scale. This last question is designed to be separate from the initial satisfaction question to prevent coupling.

4 RESULT

Fig. 3 displays A) Elevator Satisfaction (Emotion) values, B) RoSAS (Warmth/Competence/Discomfort) values, and C) Perceived Speed values for each trial obtained through the online survey. Lower Discomfort values indicate a more positive response. Error bars represent a 95% confidence interval, and significance levels are indicated as follows: * $p < .05$, ** $p < .01$, *** $p < .001$, $N = 374$. The abbreviations used in the paper represent combinations of conditions as follows: PF: Plain-Fast, PS: Plain-Slow, CF: Considerate-Fast, CS: Considerate-Slow.

4.1 Onboarding Time (Fast vs. Slow)

Regardless of the position, ANOVA analysis showed a significant difference in elevator satisfaction between Fast (PF & CF) and Slow (PS & CS) conditions ($F(1) = 24.66$; $p < .001$). When comparing the time factor for each trial, Tukey post-hoc tests revealed a significant difference between PF ($M=5.85$; $SD=1.76$) and PS ($M=5.49$; $SD=1.76$), while there was no significant difference with CS ($M=5.76$; $SD=1.89$) (PF-PS: $p < .05$ / PF-CS: $p = .87$). CF ($M=6.33$; $SD=1.77$) showed significant differences compared to PS and CS (CF-PS: $p < .001$ / CF-CS: $p < .001$).

Based on these results, we can conclude that the null hypothesis H_{A0} has been rejected, and it can be stated that a robot that boards quickly and reduces boarding time increases elevator satisfaction among passengers.

4.2 Positions & Path (Plain vs. Considerate)

Regardless of the time factor, ANOVA analysis revealed a significant difference in elevator satisfaction between the Plain (PF & PS) and Considerate (CF & CS) paths ($F(1) = 15.89$; $p < .001$). When comparing the paths within the same time condition using Tukey post-hoc tests, in the fast condition, there was a significant difference in satisfaction between the Plain path, with a mean of 5.85 ($SD=1.76$), and the Considerate path, with a mean of 6.33 ($SD=1.77$) ($p < .01$). In the slow condition, while there was a difference between the Plain path, with a mean of 5.49 ($SD=1.76$), and the Considerate path,

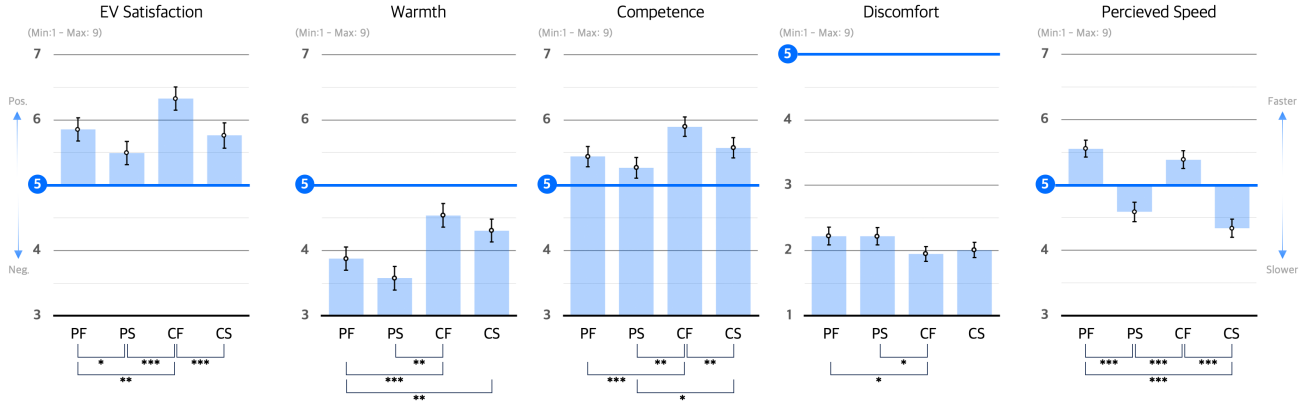


Figure 3: Survey Results for EV Satisfaction(Emotions), RoSAS, and Perceived Speed in Each Trial(PF: Plain-Fast / PS: Plain-Slow / CF: Considerate-Fast / CS: Considerate-Slow).

with a mean of 5.76 (SD=1.89), this difference was not statistically significant ($p = .17$). This suggests that the impact of the position on satisfaction is relatively reduced when the robot moves slowly. Based on these results, we can conclude that the null hypothesis H_{B0} has been rejected, and it can be stated that the position suggested in [5], which is more considerate, leads to higher satisfaction among people than the robot simply moving to the center. There was no significant difference between PF and CS ($p = .90$).

4.3 The Robotic Social Attributes Scale (RoSAS)

4.3.1 Speed - RoSAS. The ANOVA test revealed that there were significant differences in two of the three dimensions of Robotic Social Attribute Scale(RoSAS), Warmth and Competence, due to changes in speed (Warmth: $F(1) = 8.52$; $p < .01$ / Competence: $F(1) = 9.91$; $p < .01$ / Discomfort: $F(1) = 0.19$; $p = .66$). When examining the differences in RoSAS values based on speed within the same position condition using Tukey post-hoc tests, there were no significant differences in both Warmth and Competence when comparing PF and PS (Warmth: $t(1492) = 2.32$; $p = .09$ / Competence: $t(1492) = 1.55$; $p = .41$ / Discomfort: $t(1492) = 0.04$; $p = 1.00$). However, when comparing CF and CS, only Competence showed a significant difference (Warmth: $t(1492) = 1.81$; $p = .27$ / Competence: $t(1492) = 2.90$; $p < .05$ / Discomfort: $t(1492) = -0.67$ $p = .91$).

4.3.2 Position - RoSAS. ANOVA test revealed significant differences in all three dimensions of RoSAS indicators due to position changes (Warmth: $F(1) = 58.63$; $p < .001$ / Competence: $F(1) = 23.38$; $p < .001$ / Discomfort: $F(1) = 14.14$; $p < .001$). Tukey post-hoc tests showed that there were significant differences in RoSAS values when comparing PF and CF under the same speed conditions (Warmth: $t(1492) = -5.16$; $p < .001$ / Competence: $t(1492) = -4.10$; $p < .001$ / Discomfort: $t(1492) = 3.02$; $p < .001$). When comparing PS and CS, significant differences were observed only in Warmth and Competence (Warmth: $t(1492) = -1.73$; $p < .001$ / Competence: $t(1492) = -2.74$; $p < .05$ / Discomfort: $t(1492) = 2.09$ $p = .10$).

4.3.3 RoSAS - EV Satisfaction. To investigate the relationship between the robot's sociability and elevator usage satisfaction, specifically, whether higher sociability leads to increased satisfaction, Pearson's correlation analysis was conducted. The analysis revealed that for the three dimensions of RoSAS, Competence had a high positive correlation, while Warmth had a moderate positive correlation with satisfaction (Satisfaction-Warmth: $r(1494) = .43$; $p < .001$ / Competence: $r(1494) = .56$; $p < .001$ / Discomfort: $r(1494) = -.025$; $p < .001$). In the Simple Linear Regression analysis, it was found that Competence had the most significant impact on satisfaction increase (Warmth: $\beta = 0.43$; $p < .001$ / Competence: $\beta = 0.66$; $p < .001$ / Discomfort: $\beta = -0.36$; $p < .001$).

4.4 Perceived Speed

For the Fast condition, participants reported that the robot was slightly faster than expected, while for the Slow condition, it was reported to be slightly slower than expected (PF: $M=5.56$; $SD=1.29$ / CF: $M=5.39$; $SD=1.34$ / PS: $M=4.59$; $SD=1.46$ / CS: $M=4.34$; $SD=1.35$). Following ANOVA analysis, Tukey post-hoc tests revealed significant differences in speed within the same location(PF-PS: $p < .001$ / CF-CS: $p < .001$), and there were also significant differences in speed when considering different locations(PF-CS: $p < .001$ / CF-PS: $p < .001$). However, there were no significant differences in perceived speed due to position when speed remained constant (PF-CF: $p = .33$ / PS-CS: $p = .06$).

Just as the difference in robot speed (time) significantly affected satisfaction, when analyzing the correlation between the perceived speed by individuals and satisfaction using Pearson's r correlation analysis, it was found that the correlation, although not very strong, was present ($r(1494) = .37$; $p < .001$). Simple Linear Regression analysis revealed that perceived speed significantly predicted elevator satisfaction ($\beta = 0.46$; $p < .001$).

4.5 Qualitative Findings

For each trial, we collected user opinions through open-ended questions, classifying their responses into categories such as Position,

Path, Speed, Entertaining, Concern, Neutral, and others. These categories were further grouped into positive, negative, and neutral feedback. Commonly mentioned positive responses included the perception that the robot waits for people to board before itself, which was seen as considerate. Many respondents also had a positive perception of the robot, describing it as a "companion" or something they were already familiar with. Additionally, respondents appreciated the robot's hard work, cuteness, and charm. Regarding robot speed, across all trials, there was a consistent desire for the robot to be faster, with some negative feedback specifically related to waiting for the robot to complete rotations.

For the "Plain path" condition, positive opinions centered around the predictability of a straight path and the perception of minimal delay. Negative feedback often related to the central location of the robot, which obstructed other people's movements. Notably, the "Plain-Slow" (PS) condition generated the highest proportion of negative opinions, mainly concerning both position and speed.

Conversely, the "Considerate path" condition received comments expressing that the robot seemed considerate of people and appeared smarter in finding empty spaces. However, in the case of "Considerate-Slow" (CS), negative opinions regarding slow speed, such as "no need to stand in the corner if it moves slowly," were collected more than twice as much as positive opinions about the position.

5 DISCUSSION

The results of this study demonstrate the impact of robot boarding time and standing positions on elevator experience satisfaction when using elevators with robots. Additionally, this research examined the relationship between these two factors and the social acceptance of robots. In all four trials, participants' satisfaction scores exceeded 5, indicating a generally positive elevator experience with robots. However, investigating the differences between the conditions can help us understand the dynamics at play and how to enhance people's experience.

The study confirmed that boarding an elevator with a robot that moves quickly leads to higher elevator satisfaction compared to boarding with a robot that moves relatively slowly. In other words, reducing the robot's boarding and exiting time is crucial in elevator situations. In particular, the Considerate path, involving more rotations, was perceived as relatively slower compared to Plain path. Even if this difference was not statistically significant in all cases (PF-CF: Mean diff. = 0.17; $p = .33$ / PS-CS: Mean diff. = 0.25; $p = .06$), we should consider how to, not only reduce the real time required, but also how this time is perceived. Moreover, in the open-ended responses, there was a notable presence of negative feedback regarding slow robot speed, and since negative experiences tend to leave a stronger impression [13], it underscores the importance of minimizing this negative factor. At the same time, robots positioned more considerately also received higher RoSAS scores and increased elevator satisfaction.

Considering these findings, we could conclude that the Considerate-Fast condition is the best alternative, as it minimizes boarding time while respecting the most acceptable position. However, it's important to acknowledge that, in real-life scenarios, it would be

challenging for a robot to move under these conditions due to various factors. The Considerate path shown in the video, while not covering a long distance, involved multiple turns, which could result in a longer duration than when taking a straight path at the same speed (as in Plain condition). Furthermore, from a technical perspective, the Considerate path could lead to blind spots in the robot's sensing capabilities, as it might not be able to detect obstacles on its path before boarding. In other words, in practical terms, the robot would require a minimum of 16 seconds to navigate the Considerate path, equivalent to the time taken in the Slow condition. Therefore, when considering a realistic option for implementation with our current technology, a choice should be made between Plain-Fast and Considerate-Slow options. While we observed a significant for Warmth between the two trials (Warmth: $t(1492) = -3.35$; $p < .001$), we did not find significant differences for Competence and Discomfort (Competence: $t(1492) = -1.19$; $p = .63$ / Discomfort: $t(1492) = 2.34$ $p = .09$). Similarly, there was no significant difference in satisfaction ($p = .90$). So, when faced with a choice between the two, opting for Plain-Fast might be the most practical option, prioritising a reduced boarding time over the preferred position. This can reduce the potential risk associated with increased travel time in Considerate-Slow and address technical challenges, such as avoiding blind spots and sensing the interior of the elevator.

We should highlight that, in general, Competence had the most significant impact on increasing satisfaction. So, focusing on improving perceived competence would be the most impactful strategy for further developments.

6 CONCLUSION AND FUTURE WORK

The primary goal when robots share elevators with people is to enhance elevator satisfaction. Our study underscores that the most influential factor in this satisfaction is the perceived competence of the robots. People feel more satisfied with robots in elevators when they think the robot is competent. Further research is needed to identify the elements contributing to this perception.

Moreover, we've determined that both boarding time and position significantly impact satisfaction. When faced with practical implementation, choices between these two factors must be made. In our case, prioritizing the reduction of boarding time appears to be the more pragmatic approach. Simultaneously, there's an opportunity to further minimize this time.

To expedite the process and enhance the perception of competence, robots should exhibit swift movement and make efficient decisions, including improvements in communication methods. Optimizing elevator door control methods is crucial. People often commence their movements before the elevator doors are fully open, and robots should adopt a similar practice. Initiating door closure before assuming a specific position or making a turn can effectively reduce waiting time, leading to heightened passenger satisfaction.

As we continue to explore human-robot interactions, these insights and ongoing research pave the way for a more user-friendly future. Robots will increasingly contribute to our daily lives, enhancing efficiency and convenience.

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